

Math 10B with Professor Stankova

Quiz 3; Tuesday, 2/6/2018

Section #211; Time: 11 AM

GSI name: Roy Zhao

Name: _____

Circle True or False or leave blank. (1 point for correct answer, -1 for incorrect answer, 0 if left blank)

1. True **FALSE** The number of ways there are to flip 2 heads out of 5 total flips is $P(5, 2)$ because the order of the coin flips matters.

Solution: We say order doesn't matter to mean that when picking the **location** of the H 's, the order in which we pick them does not matter and hence we should use C . So order always has to do with the order in which you choose things, not in the ordering compared to everything else.

2. **TRUE** False It is impossible to devise an algorithm to solve the stable marriage problem if men can marry other men (the "roommate problem").

Solution: Zvezda gave an example of a system of preferences such that there is no stable pairing which means that an algorithm cannot give you a stable pairing.

Show your work and justify your answers. Please circle or box your final answer.

3. (10 points) (a) (4 points) How many ways can I buy 200 bubble teas from TeaOne for an event if there are 10 different options to choose from?

Solution: There are 200 indistinguishable balls which are the bubble teas, and the boxes are the options. So there are $\binom{200+10-1}{200} = \binom{209}{200}$ ways to do this.

- (b) (4 points) The most popular option is original milk tea. How many ways can I do this if I need at least 20 of that option and at least 10 of every other option?

Solution: First I buy 20 of that option and 10 of all the other options. So I've bought 110 bobas and need to buy 90 more out of the 10 options. There are $\binom{90+10-1}{90} \binom{99}{90}$ ways to do this.

- (c) (2 points) Suppose men and women have the preferences $m_1 : w_1 > w_3 > w_2$, $m_2 : w_3 > w_1 > w_2$, $m_3 : w_3 > w_2 > w_1$ and $w_1 : m_1 > m_2 > m_3$, $w_2 : m_2 > m_3 > m_1$, $w_3 : m_3 > m_2 > m_1$. Is the matching $(m_1, w_1), (m_2, w_3), (m_3, w_2)$ stable?

Solution: This is not stable because m_3 prefers w_3 to w_2 and w_3 prefers m_3 to m_2 which means that both will divorce their partners and get with each other.